

ASSIGNMENT 1

Data Mining and Retrieval

Kevin Ruberwa (101190)

Question 4: Classification, Regression, and Clustering

Classification

Classification is a supervised learning technique in which the response variable is categorical. The objective is to assign a new observation to one of the predefined classes.

Example 1: Loan Default Prediction

- **Response variable:** Default / No Default
- **Predictors:**
 - Income
 - Age
 - Loan amount
 - Debt-to-income ratio
 - Previous arrears
 - Credit history
- **Goal:** Prediction

The objective is to predict whether a new customer is likely to default on a loan based on the customer's characteristics and credit history.

Example 2: Fraud Detection

- **Response variable:** Fraud / Not Fraud
- **Predictors:**
 - Transaction amount
 - Transaction location
 - Transaction time
 - Transaction frequency
 - Merchant type
 - Customer's historical transaction behavior
- **Goal:** Prediction

The objective is to determine whether a new transaction is likely to be fraudulent.

Example 3: Email Spam Detection

- **Response variable:** Spam / Not Spam
- **Predictors:**
 - Words used in the email
 - Sender information
 - Number of links
 - Suspicious phrases
 - Attachment type
- **Goal:** Prediction

The objective is to classify a new email as either spam or not spam.

Regression

Regression is a supervised learning technique in which the response variable is numerical. It is used to estimate or predict a continuous outcome.

Example 1: House Price Prediction

- **Response variable:** House price in RWF
- **Predictors:**
 - House size
 - Number of rooms
 - Location
 - Age of the house
 - Land size
- **Goal:** Prediction

The objective is to predict the price of a new house based on its characteristics.

Example 2: Loan Loss or Credit Exposure Prediction

- **Response variable:** Expected loan loss or exposure amount
- **Predictors:**
 - Loan size
 - Income
 - Collateral
 - Loan maturity
 - Interest rate
 - Borrower characteristics
- **Goal:** Prediction

The objective is to estimate the expected loan loss or exposure amount based on the characteristics of the borrower and the loan.

Example 3: Monthly Sales Prediction

- **Response variable:** Sales revenue in RWF
- **Predictors:**
 - Advertising expenditure
 - Previous sales
 - Number of customers
 - Season
 - Product price
- **Goal:** Prediction

The objective is to predict the sales revenue for the following month using historical and business-related variables.

Regression for Inference

Regression is not limited to prediction. It can also be used for **inference**, where the objective is to understand the relationship between variables.

For example, instead of asking:

”What will next month’s sales be?”

we may ask:

”Does an increase in advertising expenditure lead to an increase in sales?”

In this case, the main objective is to understand the relationship or effect of one variable on another, rather than simply making predictions.

Therefore, **regression describes the statistical/modeling method, while prediction and inference describe the objective of the analysis.**

Clustering

Clustering is an unsupervised learning technique that groups observations according to their similarities. Unlike classification, there is no predefined response variable or class label.

Example 1: Bank Customer Segmentation

Customers can be grouped using variables such as:

- Income
- Number of transactions
- Account balances
- Loan usage
- Age

The resulting clusters may represent groups such as retail customers, high-value customers, SME-like customers, or low-activity customers. These groups can help a bank develop different products and marketing strategies.

Example 2: Credit Portfolio Segmentation

Loans can be clustered based on:

- Economic sector
- Loan exposure
- Maturity
- Interest rate
- Collateral
- Arrears behavior

This can help identify groups of loans with similar risk characteristics and support credit risk management.

Example 3: Customer Transaction Behavior

Customers can also be grouped according to their transaction behavior using variables such as:

- Number of transactions
- Average transaction amount
- ATM usage
- Mobile banking usage
- International transactions

The resulting clusters can identify different customer behavioral patterns and help the bank design more targeted services.

Question 7: K-Nearest Neighbors (KNN)

The K-Nearest Neighbors (KNN) algorithm classifies a new observation according to the classes of its nearest observations. The classification is based on a distance measure, with Euclidean distance being one of the most commonly used measures.

The given dataset is:

Observation	X_1	X_2	X_3	Class (Y)
1	0	3	0	Red
2	2	0	0	Red
3	0	1	3	Red
4	0	1	2	Green
5	-1	0	1	Green
6	1	1	1	Red

The new observation to be classified is:

$$(X_1, X_2, X_3) = (0, 0, 0)$$

Calculate the Distances

For three-dimensional observations, the Euclidean distance between a point (x_1, x_2, x_3) and the new observation $(0, 0, 0)$ is:

$$d = \sqrt{(x_1 - 0)^2 + (x_2 - 0)^2 + (x_3 - 0)^2}$$

Observation 1

$$d_1 = \sqrt{(0 - 0)^2 + (3 - 0)^2 + (0 - 0)^2} = \sqrt{9} = 3.00$$

Observation 2

$$d_2 = \sqrt{(2 - 0)^2 + (0 - 0)^2 + (0 - 0)^2} = \sqrt{4} = 2.00$$

Observation 3

$$d_3 = \sqrt{(0 - 0)^2 + (1 - 0)^2 + (3 - 0)^2} = \sqrt{10} \approx 3.16$$

Observation 4

$$d_4 = \sqrt{(0 - 0)^2 + (1 - 0)^2 + (2 - 0)^2} = \sqrt{5} \approx 2.24$$

Observation 5

$$d_5 = \sqrt{(-1 - 0)^2 + (0 - 0)^2 + (1 - 0)^2} = \sqrt{2} \approx 1.41$$

Observation 6

$$d_6 = \sqrt{(1 - 0)^2 + (1 - 0)^2 + (1 - 0)^2} = \sqrt{3} \approx 1.73$$

Therefore, the distances are:

Observation	Class	Distance
1	Red	3.00
2	Red	2.00
3	Red	3.16
4	Green	2.24
5	Green	1.41
6	Red	1.73

When ranked from the closest to the farthest:

Rank	Observation	Class	Distance
1	5	Green	1.41
2	6	Red	1.73
3	2	Red	2.00
4	4	Green	2.24
5	1	Red	3.00
6	3	Red	3.16

Classification when $K = 1$

When $K = 1$, KNN uses only the single closest observation.

The closest observation is **Observation 5**, with a distance of approximately **1.41**.

Observation 5 belongs to the **Green** class.

Therefore:

$$Y = \text{Green}$$

Thus, when $K = 1$, the new observation is classified as **Green**.

Classification when $K = 3$

When $K = 3$, the three closest observations are:

1. Observation 5 $\rightarrow$ Green
2. Observation 6 $\rightarrow$ Red
3. Observation 2 $\rightarrow$ Red

The three nearest neighbors therefore contain:

- **Red** = 2 observations
- **Green** = 1 observation

Since Red is the majority class, the new observation is classified as:

$$Y = \text{Red}$$

Therefore, when $K = 3$, the predicted class is **Red**.

Choosing K when the Bayes Decision Boundary is Highly Non-linear

If the Bayes decision boundary is highly nonlinear, a **small value of K** is generally preferred.

A small K allows KNN to focus on the local neighborhood of each observation. This makes the model more flexible and allows it to capture complex and nonlinear patterns in the data.

For example, when $K = 1$, the classification depends only on the closest observation. Therefore, the model can adapt closely to local changes in the decision boundary.

In contrast, a large K considers many observations when making a classification. This produces a smoother decision boundary because local variations are averaged out. As a result, the model may fail to capture a highly nonlinear or complex decision boundary.

Therefore:

Highly nonlinear Bayes decision boundary $\rightarrow$ Small K

However, K should not necessarily be chosen as small as possible. In practice, the value of K is usually selected using a validation procedure or cross-validation to obtain a suitable balance between model flexibility and generalization.

Question 4: Linear vs. Cubic Regression

Suppose there are 100 observations with one predictor (X) and one response (Y).

Two models are fitted:

Linear regression

$$Y = \beta_0 + \beta_1 X + \epsilon$$

Cubic regression

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3 X^3 + \epsilon$$

The cubic model is more flexible because it contains additional terms, X^2 and X^3 . The question examines how this additional flexibility affects the **training RSS** and **test RSS** under different assumptions about the true relationship between X and Y .

(a) True Relationship is Linear - Training RSS

If the true relationship between X and Y is linear, the linear regression model has the correct functional form.

However, the cubic model includes the linear model as a special case. If

$$\beta_2 = \beta_3 = 0$$

the cubic model becomes a linear model.

Because the cubic model has additional flexibility, it can fit the training observations at least as well as the linear model.

Answer:

$\text{Cubic training RSS} \leq \text{Linear training RSS}$

The cubic model may have a lower training RSS because it has more flexibility to fit the observed data.

(b) True Relationship is Linear - Test RSS

For test observations, the situation is different.

If the true relationship is linear, the linear model is correctly specified. The additional flexibility of the cubic model is unnecessary and may cause it to fit random noise in the training data.

This can lead to **overfitting**.

Therefore, the linear model is expected to have a lower test RSS.

Answer:

$\text{Linear test RSS is expected to be lower}$
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The word **expected** is important because this is not guaranteed for every particular sample. It describes the expected performance when the true relationship is linear.

(c) True Relationship is Not Linear - Training RSS

Regardless of the exact form of the nonlinear relationship, the cubic model is more flexible than the linear model.

Since both models are fitted to the same training data, the more flexible cubic model can fit the training observations at least as well as the linear model.

Answer:

$$\boxed{\text{Cubic training RSS} \leq \text{Linear training RSS}}$$

The cubic model will therefore have a training RSS that is lower than or equal to that of the linear model.

(d) True Relationship is Not Linear - Test RSS

In this case, there is not enough information to determine which model will have the lower test RSS.

If the true relationship is only slightly nonlinear, the linear model may still perform well and could have a lower test RSS.

However, if the true relationship is strongly nonlinear and can be represented well by a cubic function, the cubic model may perform better.

Answer:

$$\boxed{\text{Not enough information}}$$

The best model depends on the actual form of the nonlinear relationship.